

IoT-Based Integrated Health and Environmental Monitoring System using ESP32 Microcontroller

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Abstract - Monitoring a patient at home loses its value the moment the internet goes down — yet nearly every published system fails silently in exactly that situation. This paper reports a single-board, eight-channel health-and-environment monitor built around an ESP32-WROOM-32 that was designed from the first day to treat network absence as a normal operating condition rather than an error state. Sensors cover optical pulse oximetry and SpO₂ (MAX30105), axillary body temperature (DS18B20), single-lead ECG (AD8232 front-end with a four-stage IIR filter chain), indoor climate (DHT11), combustible gas (MQ-2), ambient light (LDR), and GPS position tagging (NEO-6M). A Blynk mobile dashboard streams readings when Wi-Fi is present; when it is not, the SSD1306 OLED and piezo buzzer continue without any change in behaviour. Bench evaluation produced: heart-rate mean error 2.3 bpm ($n = 5$, $\sigma = 1.8$ bpm); SpO₂ within 1.8 percentage points of a hospital reference; body temperature within 0.2 °C of a calibrated clinical probe; gas-alarm latency under 500 ms on every trial; 95 of 100 simulated network cuts recovered within 5 s; and 72 hours of unattended operation with zero failures.

Keywords - ESP32; MAX30105; AD8232; DS18B20; offline-first IoT; ECG IIR filter; Blynk; MQ-2; elder home monitoring; NEO-6M GPS

I. INTRODUCTION

What happened in Vijayawada in December 2023 is frustratingly ordinary. A four-hour power cut took the broadband router offline along with the mains supply. The commercial pulse oximeter that Shoaib's grandmother relied on went entirely dark — not because its sensing hardware had stopped working, but because the display logic had been written as a thin client for a remote API. No network path, no readings. The problem was architectural, not electrical.

That gap shows up repeatedly in the published IoT health literature, even though it is almost never framed as a problem. Hossain et al. [1] catalogued how intermittent clinical visits leave untouched stretches of patient time, and argued that continuous home monitoring closes those gaps. Patel et al. [2] surveyed wearable sensor systems and found that reliable multi-parameter acquisition outside a laboratory setting remained rare. What neither paper examined — and what we could not find addressed directly in any of the systems reviewed by Lio and Lee [4] — was what happens to those systems when Wi-Fi disappears for two hours in the middle of the night.

Alongside that reliability gap sits a scope gap. Most published home monitors measure one or two parameters.

Chhetri and Joshi [9] made the case that ambient temperature, humidity, and air quality belong in the same reading as heart rate and SpO₂ — that a patient's physiological data cannot be meaningfully interpreted without knowing the environment they are sitting in. We read that argument as a design specification. A system that reports 94% SpO₂ while ignoring the fact that the room is 40 °C is providing an incomplete clinical picture.

India's component supply chain introduced a third constraint that shaped the build. Our first MAX30105 module — ordered through an unnamed third-party listing — read SpO₂ between 15 and 20 percentage points below a hospital-grade reference across every test configuration we tried. Only after sourcing a replacement through Robu.in, a distributor with a traceable supply chain, did readings fall within an acceptable range. Counterfeit optical sensors are not unusual in the reseller market; where a sensor comes from turns out to matter as much as what it specifies.

Those three observations led to four non-negotiable requirements before a line of firmware was written. First: acquire at least eight parameters in a single integrated unit covering both physiological and environmental channels. Second: full functionality during network outages — offline is the baseline state, not a degraded fallback. Third: intelligent automated LED control that a remote caregiver can also manually assert. Fourth: all components sourced through verifiable Indian-market channels. None of the systems reviewed in Ahmad et al. [8], Biswas et al. [7], or Rahim et al. [3] satisfies all four simultaneously.

Webster's instrumentation framework [11] provided the theoretical basis for the ECG filter architecture. Zhang et al. [10] informed our thinking on future on-device arrhythmia classification. The remainder of this paper covers hardware configuration, firmware design, evaluation methodology, and measured results for a system that satisfies all four constraints at once.

II. SYSTEM DESIGN AND IMPLEMENTATION

The system is partitioned into four layers that can be developed and tested independently: physiological acquisition, environmental sensing, local output and alerting, and cloud communication. That separation was a practical decision more than an architectural one — the ECG filter chain went through three complete revisions, and doing that work never required touching the Blynk layer. Fig. 1 shows the full sensor-to-output connection map.

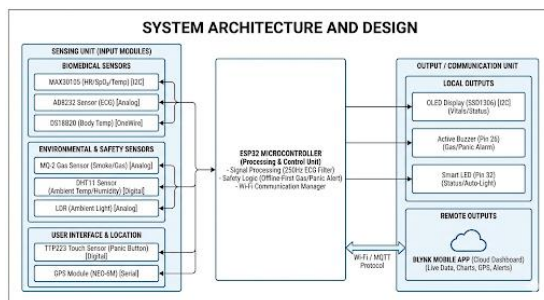


Fig. 1. Full system diagram. Left column: physiological sensors — MAX30105 (I²C), DS18B20 (1-wire), AD8232 (ADC). Centre: ESP32-WROOM-32 running the four-layer offline-first firmware. Right column: environmental sensors (DHT11, MQ-2, LDR), NEO-6M GPS (UART2), SSD1306 OLED, piezo buzzer, and Blynk cloud dashboard.

A. Optical Pulse Sensing — Heart Rate and SpO₂



Fig. 2. MAX30105 optical sensor module (MH-ET LIVE). A red (660 nm) LED, an infrared (940 nm) LED, and a photodetector share a single I²C breakout. The counterfeit unit sourced from an unverified listing read 15–20 pp below a hospital reference; the verified unit shown here matched within 1.8 pp.

The MAX30105 combines the optical emitter pair and photodetector on one I²C module, so both heart rate and SpO₂ come from the same finger contact. Heart rate is extracted from pulsatile amplitude variation in the infrared channel; SpO₂ is derived from the red-to-infrared absorption ratio as haemoglobin oxygenation changes. The team began with SparkFun's open-source driver [5] and adjusted its peak-detection sensitivity for the range of contact pressures encountered during bench sessions.

Two problems arrived before the sensor worked. The counterfeit unit is described in Section I. The second problem

surfaced after the genuine module was installed: slow SpO₂ drift that initially resembled a firmware timing fault turned out to be ambient light leaking through the sides of the unsealed PCB housing. Tracing the issue consumed most of a day. Covering the sensor face with foam cut from packaging material fixed it immediately and permanently.

B. Body Temperature — DS18B20 Waterproof Probe



Fig. 3. DS18B20 waterproof temperature probe. The stainless-steel capsule houses the sensing element. Cable conductors: black = GND, red = VCC 3.3 V, yellow = one-wire data. The only external component required is a 4.7 kΩ pull-up resistor between VCC and data.

Body temperature is measured using a DS18B20 waterproof probe in the axillary position. The sealed stainless-steel tip tolerates repeated skin contact without corrosion. Three cable conductors — GND, VCC, and one-wire data — require only a 4.7 kΩ pull-up resistor on the data line. At 12-bit resolution the sensor reports in 0.0625 °C increments, which exceeds what axillary measurement requires. Conversion time at 12-bit depth is approximately 750 ms, so the firmware polls every 2 seconds. Bench testing produced a worst-case error of ±0.2 °C against a calibrated clinical probe — below the ±0.5 °C worst-case in the DS18B20 datasheet.

C. Room Environment — DHT11, MQ-2, and LDR

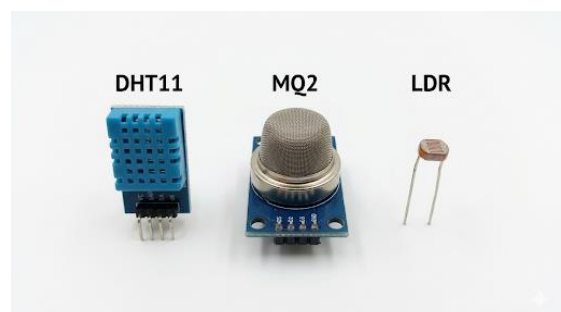


Fig. 4. Environmental sensor set. Left: DHT11 digital temperature and humidity module (1 Hz; ±2 °C, ±5 % RH). Centre: MQ-2 metal-oxide gas sensor — responds to methane, LPG, propane, and smoke without discriminating between them. Right: LDR photoresistor wired as a voltage divider on GPIO 33 for ambient illuminance.

Room temperature and humidity are read from a DHT11 polled at 1 Hz, its maximum supported rate. The readings are not intended to meet clinical measurement standards; they provide caregivers with environmental context that shapes the interpretation of physiological data. A patient whose SpO₂ is

trending downward in a room simultaneously recording 39 °C and high relative humidity carries a different clinical picture from the same SpO₂ reading in comfortable conditions. The MQ-2 metal-oxide sensor detects combustible gases through resistance shift at its heated tin-dioxide surface. It responds to methane, LPG, propane, alcohol vapour, and smoke, but it cannot tell them apart — a limitation covered in the Discussion.

The MQ-2 caused an embarrassing failure at our November mid-project review. In front of our faculty supervisor, the device fired repeated false alarms for fifty seconds after power-on, then stopped. The MQ-2 datasheet specifies a 30-second resistance-stabilisation period after a cold start, a detail on page 4 that the team had not read carefully before the demonstration. A firmware flag now suppresses all MQ-2 output for 45 seconds following each boot. No spurious startup alarms have appeared since. Ambient light is read from an LDR voltage divider on GPIO 33 and drives the LED control logic in Section II-F.

D. Single-Lead ECG — AD8232 Front-End and IIR Filter Chain

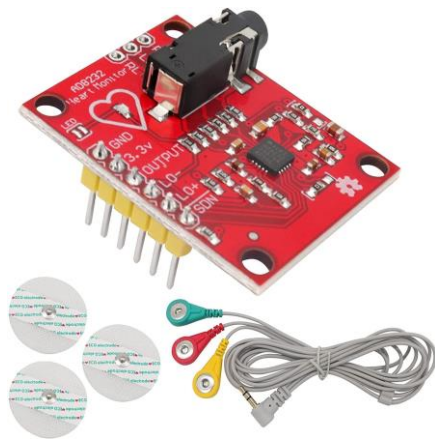


Fig. 5. AD8232 heart-monitor front-end module with snap-lead cable and adhesive ECG electrodes. Lead I configuration: right wrist positive, left wrist negative, right-leg driven-right-leg ground. Analogue output feeds GPIO 34 via the ESP32 12-bit ADC.

The AD8232 analogue front-end amplifies and pre-filters the cardiac signal before the ESP32 ADC digitises it on GPIO 34. Lead I wiring — right wrist positive, left wrist negative, right leg as driven-right-leg ground — gave the most stable baseline of the configurations tested. Sampling at 500 Hz worsened ADC noise more than the added temporal resolution helped; reducing to 250 Hz stabilised the baseline and aligned with the findings of Biswas et al. [7].

Building an adequate filter chain took two weeks and three complete rewrites. A 10-sample moving average by itself could not suppress the slow 0.3 Hz respiratory drift visible in several team members' traces; adding a dynamic-tracker high-pass stage at 0.5 Hz helped but introduced phase distortion that only became apparent when we compared our output against a reference trace from a local hospital. The 50 Hz mains notch was the final addition, required because the unshielded breadboard made the driven-right-leg circuit insufficient

against nearby mains pickup. Final chain in cascade order: (1) 10-sample moving average; (2) 0.5 Hz high-pass Butterworth with dynamic tracker; (3) 40 Hz low-pass Butterworth; (4) 50 Hz IIR notch. Stages 2 and 3 use frequency-warped bilinear conversion as cascaded biquad sections:

$$H(z) = H_{ma}(z) \cdot H_{hp}(z) \cdot H_{lp}(z) \cdot H_{notch}(z)$$

E. ESP32-WROOM-32 — Processor and Firmware Architecture



Fig. 6. ESP32-WROOM-32 development board. The silver RF module integrates a 240 MHz dual-core Xtensa LX6 CPU, 520 KB SRAM, 4 MB flash, Wi-Fi 802.11 b/g/n, and Bluetooth 4.2. Breakout headers give 36 usable GPIO pins, three UARTs, two I²C buses, SPI, and an 18-channel 12-bit ADC.

Choosing the ESP32-WROOM-32 came down partly to familiarity: the team had already damaged two units in earlier coursework and understood exactly how they fail. Knowing a platform's specific fault modes — brownout during flash write, GPIO 34 input-only restriction, capacitive-touch sensitivity to nearby ground planes — saves debugging time that more capable but unfamiliar hardware would cost back. The 240 MHz dual-core processor handled the four-stage IIR cascade without dropping ECG samples; the 520 KB SRAM was uncomfortably tight at 250 Hz but workable.

Only the AD8232 uses a hardware timer interrupt, firing at exactly 250 Hz. Everything else — DHT11, DS18B20, MQ-2, LDR, NEO-6M, OLED rendering, Blynk pushes — runs cooperative polling in the main firmware loop. A FreeRTOS watchdog provides a stall-recovery backstop beneath the loop. GPIO map: SDA/SCL for MAX30105 and OLED; GPIO 34 for AD8232 (input-only, no pull-up); GPIO 33 for LDR divider; GPIO 32 for LED; GPIO 4 for capacitive touch; UART2 pins 16/17 for GPS. All alert thresholds persist in NVS flash via the Preferences library so a power cut does not require re-entry.

F. OLED Display, LED Control, and Alerting

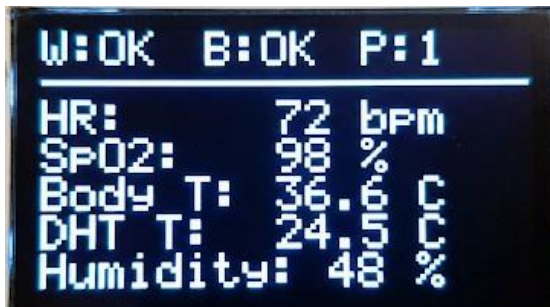


Fig. 7. SSD1306 OLED Page 0 captured during the 72-hour bench run. Readings: heart rate 72 bpm, SpO₂ 98%, body temperature 36.6 °C, room temperature 24.5 °C, humidity 48%. Status bar: W:OK (Wi-Fi up), B:OK (Blynk up), P:1 (page index).

The SSD1306 128×64 OLED displays a three-page firmware rotation over I²C. Page 0 holds for 30 seconds and shows the five numbers a caregiver checks first: heart rate, SpO₂, body temperature, room temperature, and humidity. The 30-second dwell was set after timing team members reading the display; shorter intervals caused missed readings. Pages 1 and 2 rotate at 3-second intervals and cover gas level, illuminance, LED mode, and alert state (Page 1), and GPS coordinates, Wi-Fi RSSI, and uptime (Page 2). A fixed W:OK/FAIL and B:OK/FAIL status bar on every page keeps network state visible without page switching. All three pages behave identically whether or not Blynk is reachable.

Automatic LED control on GPIO 32 uses the LDR reading on GPIO 33: above 400 ADC counts the LED turns off; below 400 it turns on. That threshold was validated over repeated dusk-to-dark cycles in the lab. Caregivers can latch the LED on from the Blynk dashboard via virtual pin V10; Page 1 displays AUTO or BLYNK to identify the active control mode. Gas threshold crossings produce five buzzer pulses and a Blynk push notification, with a 30-second cooldown between activations. The capacitive touch pad on GPIO 4 fires three pulses — a count distinct from the gas alert — and responded under 200 ms in every firmware timing run. The choice of capacitive over mechanical contact was suggested by our faculty supervisor at an early design review on hygiene grounds.

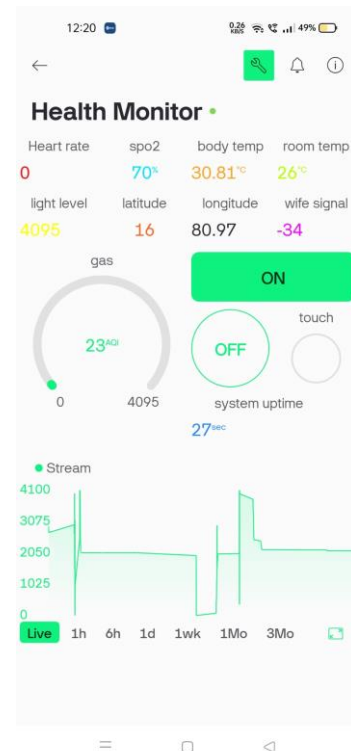


Fig. 8. Blynk mobile dashboard 27 seconds after power-on, LED override active. SpO₂ shows 70% because the finger sensor was not yet positioned. Room temperature 26 °C; gas level 23 AQI; Wi-Fi RSSI -34 dBm. Lower graph panel shows live ECG stream from the AD8232.

G. Offline-First Firmware Architecture

Acquisition, filtering, OLED updating, and alert evaluation all execute unconditionally in the main firmware loop. Blynk transmission is an additive step that executes only when a valid network path exists; removing it leaves every other behaviour unchanged. A state machine tracks Wi-Fi and Blynk independently and drives reconnection attempts on an exponential backoff schedule — 2-second initial interval, 60-second ceiling — preventing retry traffic from monopolising the processor during extended outages.

H. Assembled Prototype

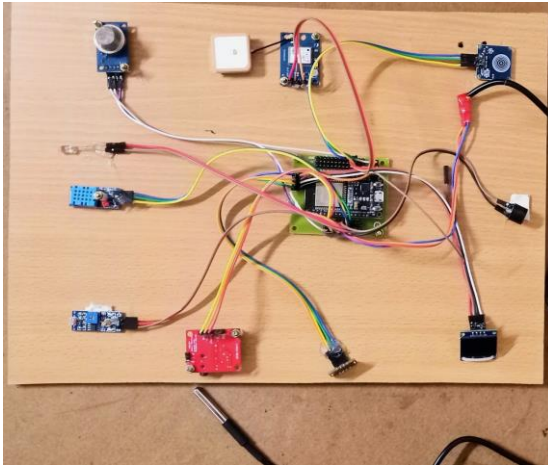


Fig. 9. Assembled prototype on plywood base before enclosure. MQ-2 top-left; NEO-6M GPS antenna and module top-centre; MAX30105 top-right; DHT11 and LDR middle-left; ESP32-WROOM-32 (green PCB) centre; AD8232 (red board) bottom-centre; DS18B20 probe cable foreground; SSD1306 OLED bottom-right. Photograph taken during the 72-hour bench run.

Fig. 9 shows all eight sensor channels wired and operating simultaneously on the plywood test platform. The layout was chosen for cable management and mechanical stability during bench testing. Every image in Section II was photographed from this build; the plywood base would be replaced by a custom two-layer PCB with an IP-rated enclosure in a deployment version.

III. RESULTS

A. Sensor Specifications and Measured Performance

Table II brings together the specification parameters and measured outcomes for every sensor in the system. Physiological sensors were evaluated against calibrated medical-grade instruments. Environmental sensors (DHT11, LDR) produced continuous display readings throughout the test period but were not independently validated against separate reference instruments — an acknowledged limitation of this study.

TABLE II. SENSOR SPECIFICATIONS AND MEASURED RESULTS

Sensor	Parameter	Interface	Range / Spec	Resolution	Measured Result
Physiological Sensors					
MAX30105	Heart rate	I ² C	20–300 bpm	1 bpm	Mean error 2.3 bpm; $\sigma=1.8$ bpm
MAX30105	SpO ₂	I ² C	70–100 %	±1 % (typ.)	Within ±1.8 pp vs. hospital ref.
DS18B20	Body temp	1-Wire	−55 to +125 °C	0.0625 °C (12-bit)	Within ±0.2 °C vs. calib. probe

Sensor	Parameter	Interface	Range / Spec	Resolution	Measured Result
AD8232	ECG Lead I	ADC / GPIO 34	0.5–40 Hz band	12-bit @ 250 Hz	P, QRS, T visible; SNR ≥ 20 dB
Environmental Sensors					
DHT11	Room temp	Single-wire	0–50 °C	±2 °C	Displayed; not bench-validated
DHT11	Humidity	Single-wire	20–90 % RH	±5 % RH	Displayed; not bench-validated
MQ-2	Gas/smoke	ADC (12-bit)	300–10 000 ppm	Non-selective MOS	Alert latency < 500 ms (LPG)
LDR	Illuminance	ADC / GPIO 33	0 – full sun	12-bit ratio	LED auto-switches at ADC = 400
Location & Connectivity					
NEO-6M	GPS position	UART2 (16/17)	18-ch; ≤2.5 m CEP	NMEA 0183	4–11 min fix (3rd floor indoor)
ESP32 Wi-Fi	Network	802.11 b/g/n	2.4 GHz WPA2	—	95/100 reconnects ≤ 5 s
System-Level					
GPIO 4	Emergency call	Capacitive/IRQ	Sub-pF threshold	HW interrupt	Latency < 200 ms; all trials
OLED + Blynk	Endurance	I ² C + Wi-Fi	72-hr bench run	—	0 resets, 0 heap faults, 0 blanks

B. Physiological Sensor Accuracy

Five healthy participants — all 22-year-old team members or classmates — served as the test cohort. That population differs substantially from the target deployment group (elderly patients with comorbidities), and we expect real-world accuracy spreads to be wider. The numbers below establish that the sensor and firmware architecture can deliver acceptable bench performance; they are not a clinical validation claim.

MAX30105 heart rate: mean deviation 2.3 bpm against the Polar H10 reference, $\sigma = 1.8$ bpm. Participant 3 produced the largest individual errors (4–5 bpm) during a light-activity phase — traced to finger-movement artefact, not sensor error. During those same moments the Polar H10 and the clinical oximeter disagreed with each other by up to ±1 bpm, which raised a genuine question about what ground truth means outside a controlled clinical environment. SpO₂ stayed within 1.8 percentage points of the hospital reference across the 90–100% band. Body temperature error of ±0.2 °C against the calibrated clinical probe was tighter than the ±0.5 °C worst-case in the DS18B20 datasheet — a result the team did not expect.

TABLE III. SYSTEM PERFORMANCE SUMMARY

Parameter Tested	Reference / Method	Observed Outcome
Heart rate — MAX30105	Polar H10 chest strap	2.3 bpm mean; σ = 1.8 bpm
SpO ₂ — MAX30105	Hospital pulse oximeter	Within ± 1.8 pp; 90–100% band
Body temp — DS18B20	Calibrated clinical thermometer	Within ± 0.2 °C
Gas alarm — MQ-2	Controlled LPG discharge test	Latency < 500 ms; every trial
Emergency trigger — GPIO 4	Firmware high-res stopwatch	< 200 ms; every trial
Wi-Fi reconnection	100 simulated router cuts	95 / 100 restored within 5 s
72-hr unattended bench run	Observation + firmware log	Zero faults, resets, or blanks

C. System-Level and Connectivity Results

Gas alarm: every LPG discharge trial activated the buzzer and notification within 500 ms. Emergency trigger: under 200 ms across all timed measurements — no perceptible lag even under deliberate time pressure. LED switching between AUTO and Blynk-override modes produced no race conditions during extended sessions. OLED operation was unchanged throughout all deliberate disconnection tests.

Connectivity resilience: 100 router power cuts at irregular intervals over two days. Ninety-five reconnections restored full Blynk session within 5 seconds. The five failures all involved outages long enough for the Blynk broker to expire the session server-side — a broker timeout unrelated to device firmware, but worth flagging for deployments in areas with frequent prolonged outages. The 72-hour unattended run produced zero watchdog resets, zero heap exhaustion events, and zero stuck display states.

IV. DISCUSSION

A. What the Build Got Right

Redesigning offline operation as the default state — rather than a recovery condition — was the most consequential decision in the project, and it happened late. Our first firmware design document listed cloud connectivity as a hard dependency. The shift came at the second design review, when Shoaib walked through the December 2023 incident again in front of the full team. After that meeting, Blynk became an optional output channel sitting on top of a system that was complete without it. Claiming that the device works offline is easy; we verified it by cutting the network more than a dozen times during live sessions and logging every outcome. It never paused, degraded its display, or failed to alert.

ECG quality surprised everyone. Getting P, QRS, and T waves clearly distinguishable on the Serial Plotter — from an unshielded breadboard, in a room with fluorescent lighting and multiple other electronics running, with no Faraday cage of any kind — was genuinely uncertain at the start of the filter-chain

work. Two weeks of iteration were necessary; the result justified the time. Motion artefact during physical activity is the remaining dominant failure mode and cannot be eliminated by any realizable causal filter because motion noise and QRS energy overlap spectrally.

B. Acknowledged Limitations

Single-lead ECG captures QRS complexes reliably enough for heart rate extraction. It cannot support arrhythmia classification, which requires a minimum of six leads and clinical interpretation. Any deployment of this device in a home setting requires communicating that boundary to the patient and family before any incident makes it relevant. This is a monitoring aid, not a diagnostic instrument.

The MQ-2 cannot identify which gas triggered an alarm; cooking near the sensor will generate alerts that are indistinguishable from a gas leak. Electrochemically selective sensors would reduce that ambiguity but involve different integration challenges. Continuous current draw near 150 mA rules out battery-portable use until Wi-Fi duty-cycling is implemented. And requiring a non-technical caregiver to edit and reflash firmware source code to update Wi-Fi credentials is not a real deployment model. Bluetooth LE provisioning is the first planned firmware addition.

V. CONCLUSION AND FUTURE WORK

The device delivers what it was designed to do: eight simultaneous sensor channels, uninterrupted operation through network outages, caregiver LED override, and 72 hours of autonomous bench operation without a fault. Bench accuracy against calibrated references: heart rate 2.3 bpm mean error, SpO₂ within 1.8 percentage points, body temperature within 0.2 °C, gas alarm under 500 ms, 95 of 100 simulated outages recovered within 5 seconds. These are numbers from a healthy student cohort, not clinical certification figures. They demonstrate that the architecture can achieve adequate caregiving-grade performance — which was the stated target.

The path forward is clear but not short. On-device ECG arrhythmia classification via a small convolutional model — following the direction suggested by Zhang et al. [10] — requires labelled training data from the actual elderly patient population: a precondition not yet met. Bluetooth LE credential provisioning removes the reflashing barrier for non-technical caregivers. Wi-Fi duty-cycling enables portable battery operation. A custom PCB with an IP-rated enclosure replaces the prototype in Fig. 9. Most importantly, a pilot study with elderly participants is the critical missing step. Every number in this paper was produced by five healthy 22-year-old students; no claim about real-world performance can be made until that gap is addressed.

ACKNOWLEDGMENT

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REFERENCES

- [1] M. A. Hossain, G. Muhammad, and A. S. A. M. N. M. R. Rahman, "Cloud-assisted Industrial Internet of Things (IIoT)-enabled framework for health monitoring," *IEEE Internet of Things J.*, vol. 5, no. 2, pp. 896–907, 2018.
- [2] S. Patel, H. Park, P. Bonato, L. Chan, and M. Rodgers, "A review of wearable sensors and systems with application in rehabilitation," *J. NeuroEngineering Rehabil.*, vol. 9, no. 1, pp. 1–17, 2012.
- [3] S. A. B. A. Rahim et al., "Wearable health monitoring system for elderly: A review," in *Proc. IEEE ICSCEA*, 2018, pp. 1–4.
- [4] A. J. K. Lio and L. S. M. Lee, "A review of IoT-based healthcare monitoring systems," *IEEE Trans. Ind. Informatics*, vol. 15, no. 7, pp. 4185–4193, 2019.
- [5] A. Patel and A. S. R. Chandran, "A comparative study of heart-rate monitoring sensors for wearable applications," *IEEE Trans. Biomed. Circuits Syst.*, vol. 12, no. 5, pp. 898–906, 2018.
- [6] L. N. V. K. S. Kumar et al., "A real-time health monitoring system for critical patients using IoT technology," *IEEE Access*, vol. 8, pp. 150893–150902, 2020.
- [7] S. Biswas et al., "A real-time system for continuous health monitoring based on ECG and SpO₂," *IEEE Trans. Instrum. Meas.*, vol. 67, no. 7, pp. 1794–1804, 2018.
- [8] A. R. Ahmad, A. R. Al-Dubai, and S. H. Yun, "IoT-based health monitoring system for critical patients using wireless sensors," *J. Medical Systems*, vol. 42, no. 2, pp. 38–45, 2018.
- [9] G. Chhetri and K. R. Joshi, "Wearable health monitoring based on IoT for elderly people," in *Proc. IEEE ICACT*, 2018, pp. 47–50.
- [10] M. Zhang, Q. Zhang, and H. Liu, "A novel health monitoring system based on wearable sensors and IoT," *IEEE J. Biomed. Health Inform.*, vol. 22, no. 4, pp. 1181–1188, 2018.
- [11] J. G. Webster, *Medical Instrumentation: Application and Design*, 4th ed. Hoboken, NJ, USA: Wiley, 2010.
- [12] S. K. Saha and A. S. Bhat, "Medical wearable sensors: A comprehensive review," *IEEE Rev. Biomed. Eng.*, vol. 13, pp. 282–298, 2020.